Template Document

Academic Support Agent

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Abstract

This is an example SIAM LaTeX article. This can be used as a template for new articles. Abstracts must be able to stand alone and so cannot contain citations to the paper's references, equations, etc. An abstract must consist of a single paragraph and be concise. Because of online formatting, abstracts must appear as plain as possible. Any equations should be inline.

example, LaTeX
68Q25, 68R10, 68U05

1 Introduction

The introduction introduces the context and summarizes the manuscript. It is importantly to clearly state the contributions of this piece of work. The next two paragraphs are text filler, generated by the `lipsum` package.

[2-3]

The paper is organized as follows. Our main results are in sec:main, our new algorithm is in sec:alg, experimental results are in sec:experiments, and the conclusions follow in sec:conclusions.

2 Main results

We interleave text filler with some example theorems and theorem-like items.

[4]

Here we state our main result as thm:bigthm; the proof is deferred to sec:proof.

[LDL^T Factorization [**?**]] If $A \in \mathbb{R}^{n\times n}$ is symmetric and the principal submatrix $A(1:k, 1:k)$ is nonsingular for $k = 1:n-1$, then there exists a unit lower triangular matrix L and a diagonal matrix

$$D = (d_1, \ldots, d_n)$$

such that $A = LDL^T$. The factorization is unique.

[6]

[Mean Value Theorem] Suppose f is a function that is continuous on the closed interval $[a, b]$. and differentiable on the open interval (a, b). Then there exists a number c such that $a < c < b$ and

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

In other words,

$$f(b) - f(a) = f'(c)(b - a).$$

Observe that thm:bigthm,thm:mvt,cor:a correctly mix references to multiple labels.

Let $f(x)$ be continuous and differentiable everywhere. If $f(x)$ has at least two roots, then $f'(x)$ must have at least one root. Let a and b be two distinct roots of f. By thm:mvt, there exists a number c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{0 - 0}{b - a} = 0.$$

Note that it may require two LaTeX compilations for the proof marks to show.

Display matrices can be rendered using environments from `amsmath`:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{1}$$

Equation eq:matrices shows some example matrices.

We calculate the Fréchet derivative of F as follows:

$$F'(U,V)(H,K) = \langle R(U,V), H\Sigma V^T + U\Sigma K^T - P(H\Sigma V^T + U\Sigma K^T)\rangle \tag{2a}$$
$$= \langle R(U,V), H\Sigma V^T + U\Sigma K^T\rangle$$
$$= \langle R(U,V)V\Sigma^T, H\rangle + \langle \Sigma^T U^T R(U,V), K^T\rangle. \tag{2b}$$

eq:aa is the first line, and eq:bb is the last line.

3 Algorithm

[40]

Our analysis leads to the algorithm in alg:buildtree.

Define $P := T := \{\{1\}, \ldots, \{d\}\}$ $\#P > 1$ Choose $C' \in \mathcal{C}_p(P)$ with $C' := \operatorname{argmin}_{C \in \mathcal{C}_p(P)} \varrho(C)$ Find an optimal partition tree $T_{C'}$ Update $P := (P \backslash C') \cup \{\bigcup_{t \in C'} t\}$ Update $T := T \cup \{\bigcup_{t \in \tau} t : \tau \in T_{C'} \backslash \mathcal{L}(T_{C'})\}$ T

[41]

4 Experimental results

[50]

fig:testfig shows some example results. Additional results are available in the supplement in tab:foo.

Figure 1: Example figure using external image files.

[51]

5 Discussion of $\boldsymbol{Z = X \cup Y}$

[76]

6 Conclusions

Some conclusions here.

7 An example appendix

[71]

Test Lemma.

Acknowledgments

We would like to acknowledge the assistance of volunteers in putting together this example manuscript and supplement.